1. A notebook costs twice as much as a pen. A notebook and two pens cost \$12 altogether. How much does one notebook cost?

   (A) \$2 **(B)** \$3 **(C)** \$4 **(D)** \$5 **(E)** \$6

   proposed by: Christopher Li

   Solution: Let x be the cost of a notebook. Then two pens cost x. Therefore, $x+x=12 \implies x=$ $\boxed{\textbf{(E)}\ 6}$.

2. The numbers $1, 2, 3, 4, 5, 6, 7, 8, 9$ are written on a board. Two numbers are erased, and the sum of the seven remaining numbers is 35. How many different pairs of numbers could have been erased?

   (A) 4 **(B)** 5 **(C)** 8 **(D)** 9 **(E)** 10

   proposed by: Christopher Li

   Solution: The sum of the two numbers erased must be $1+2+3+4+5+6+7+8+9-35 = 45-35=10$. The possible pairs are $(1,9)$, $(2,8)$, $(3,7)$, and $(4,6)$, giving $\boxed{\textbf{(A)}\ 4}$ pairs.

3. A jar contains fewer than 30 beads. When the beads are arranged in groups of 4, there are 3 beads left over. When they are arranged in groups of 5, there are 4 beads left over. How many beads are in the jar?

   (A) 9 **(B)** 11 **(C)** 14 **(D)** 15 **(E)** 19

   proposed by: Christopher Li

   Solution: Let b be the number of beads in the jar. We are given $b \equiv 3 \pmod 4$ and $b \equiv 4 \pmod 5$. Therefore, $b \equiv -1 \pmod 4$ and $b \equiv -1 \pmod 5$, which gives $b \equiv -1 \pmod{20}$. The only positive integer less than 30 satisfying this congruence is $\boxed{\textbf{(E)}\ 19}$.

4. What is the smallest possible radius of a circle that passes through the points $(0,0)$ and $(1,1)$?

   (A) $\frac{1}{2}$ **(B)** $\frac{\sqrt{2}}{2}$ **(C)** $\frac{\sqrt{3}}{2}$ **(D)** 1 **(E)** 2

   proposed by: Daniel Chen

   Solution: Let r be the radius. The maximum distance between two points on the circle is the diameter $2r$, so

   $$2r \geq \sqrt{1^2+1^2} = \sqrt{2}.$$

Thus $r \geq \sqrt{2}/2$. Equality is obtained when the line segment from $(0,0)$ to $(1,1)$ is a diameter of the circle. Therefore, the smallest possible radius is $\boxed{\textbf{(B)}\ \sqrt{2}/2}$.

5. The graphs of $y = x^2 - x + 2$ and $y = x^2 + bx + c$ have at least two distinct points in common. What is the value of $b + c$?

(A) -3 **(B)** -1 **(C)** 1 **(D)** 2 **(E)** 3

proposed by: Daniel Chen

Solution: Any point (x, y) on both graphs must satisfy $x^2 - x + 2 = x^2 + bx + c \implies x(b+1) = 2 - c$. If $b \neq -1$, then there is only one solution for x, implying there is only one point common to both graphs. Therefore, $b = -1$ and $c = 2$. Finally, $b + c = \boxed{\textbf{(C)}\ 1}$.

6. Rose writes four distinct positive integers on a blackboard. Calum notices that their mean is 4. Calum then writes an additional number on the board, and Rose notes that the median of all five numbers is 6. What is the product of the original four numbers?

(A) 16 **(B)** 42 **(C)** 48 **(D)** 80 **(E)** 84

proposed by: Daniel Chen

Solution: Write the original four numbers as $a < b < c < d$. Their sum is $4 \cdot 4 = 16$. Since the median of five numbers is 6, at least three of the five numbers must be at least 6. At least two of these numbers must be part of the original four, so $c \geq 6$ and $d \geq 7$.

Since $a \geq 1$ and $b \geq 2$, $16 = a + b + c + d \geq 1 + 2 + 6 + 7 = 16$. This forces $a = 1$, $b = 2$, $c = 6$, $d = 7$, giving $abcd = \boxed{\textbf{(E)}\ 84}$.

7. A positive integer is divisible by 36, and each digit in its decimal representation is either 0 or 1. What is the smallest possible number of digits it can have?

(A) 10 **(B)** 11 **(C)** 12 **(D)** 18 **(E)** 36

proposed by: Christopher Li

Solution: Let n be a multiple of 36 whose digits are all either 0 or 1. Since 4 divides n, 4 divides the two-digit number formed by taking the last two digits of n. Therefore, the last two digits of n are 00. Since 9 divides n, the sum of the digits of n must be divisible by 9. Thus, n has at least nine 1's and therefore at least 11 digits. Equality occurs when $n = 11111111100$. Therefore, $\boxed{\textbf{(B)}\ 11}$ is the smallest number of digits n can have.

8. Each square of a 10×10 grid is labeled with the fewest moves needed to travel from that square to the top-left square, where each move goes to a square sharing a side. What is the sum of all the labels?

(A) 90 **(B)** 450 **(C)** 900 **(D)** 1000 **(E)** 1100

proposed by: Daniel Chen

Solution: Number the rows from top to bottom and the columns from left to right, starting at 1. Then the square in row r and column c has label $(r-1)+(c-1) = r+c-2$. The labeled grid is shown below.

0	1	2	3	4	5	6	7	8	9
1	2	3	4	5	6	7	8	9	10
2	3	4	5	6	7	8	9	10	11
3	4	5	6	7	8	9	10	11	12
4	5	6	7	8	9	10	11	12	13
5	6	7	8	9	10	11	12	13	14
6	7	8	9	10	11	12	13	14	15
7	8	9	10	11	12	13	14	15	16
8	9	10	11	12	13	14	15	16	17
9	10	11	12	13	14	15	16	17	18

Pair the square in row r and column c with the square in row $11-r$ and column $11-c$. Their labels sum to

$$(r+c-2)+((11-r)+(11-c)-2) = 18.$$

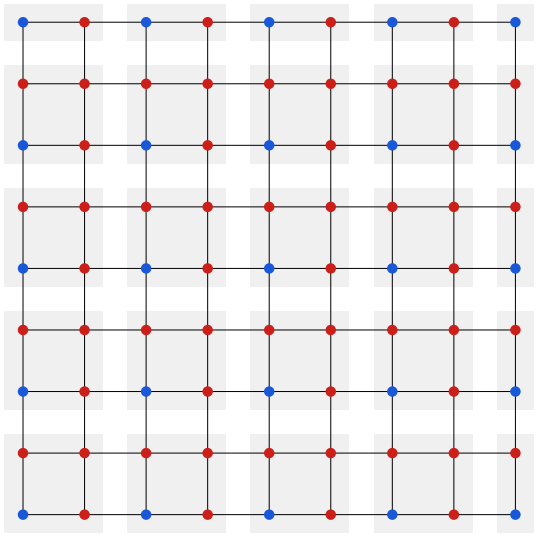
There are $100/2 = 50$ such pairs, so the sum of the labels is $50 \cdot 18 = \boxed{\textbf{(C)}\ 900}$.

9. An 8×8 square grid has $9 \times 9 = 81$ lattice points, the vertices of its unit cells. Some of these points are colored red so that every unit cell has at least three red vertices. What is the smallest possible number of red points?

(A) 25 **(B)** 48 **(C)** 52 **(D)** 56 **(E)** 61

proposed by: Terry Yang

Solution: Call an uncolored point *blue*. Every unit cell can have at most one blue vertex. Partition the 81 vertices into 25 groups as seen below.



Any two distinct vertices in the same group belong to a common unit cell, so each group can contain at most one blue point. There are $5 \cdot 5 = 25$ groups, giving at most 25 blue points. Therefore, the minimum number of red points is $81 - 25 =$ **(D)** 56.

10. The largest positive divisor of a positive integer N, other than N itself, is 35. What is the greatest possible value of N?

(A) 35 **(B)** 70 **(C)** 105 **(D)** 175 **(E)** 245

proposed by: Christopher Li

Solution: Since 35 divides N, 5 divides N. Thus $N/5$ is a positive divisor of N smaller than N, and consequently $N/5 \leq 35$. This gives $N \leq 175$.

The divisors of 175 are $1, 5, 7, 25, 35, 175$, so 175 satisfies the condition. The greatest possible value of N is **(D)** 175.

11. Let ω be the unit circle centered at the origin. Initially, one point P is marked on ω. More points are then marked by repeatedly applying the following rules until no new points can be added:

i. If a point on ω lies on the same vertical line as a marked point, mark it.

ii. If a point on ω lies on the same horizontal line as a marked point, mark it.

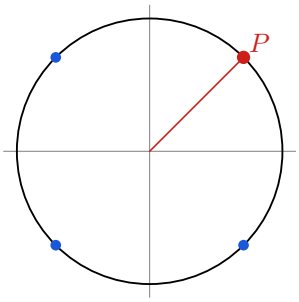
iii. If distinct points $A, B \in \omega$ satisfy that B is marked and line AB has slope 1, mark A.

What is the sum of the maximum and minimum possible numbers of distinct marked points, over all choices of the initial point P?

(A) 6 **(B)** 8 **(C)** 10 **(D)** 12 **(E)** 16

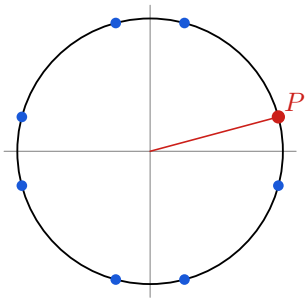
proposed by: Daniel Chen

Solution: Starting at angle θ, the rules produce $\pm\theta$ plus multiples of $90°$. This gives at least four and at most eight points, as shown below:



Minimum: 4 points

$\theta = 45°$



Maximum: 8 points

$\theta = 15°$

Therefore, the sum is $4 + 8 =$ **(D)** 12.

12. Four or more distinct prime numbers are chosen. The product of each pair of chosen primes is calculated, and these products are added. What is the smallest possible sum that is divisible by 3?

(A) 21 **(B)** 87 **(C)** 141 **(D)** 210 **(E)** 213

proposed by: Terry Yang

Solution: For a set of four or more distinct primes, let its *score* be the sum of the pairwise products. Given a set of primes, increasing any chosen prime or adding another prime increases the score. If at least five primes are chosen, the score is at least the score for $2, 3, 5, 7, 11$, which is 288.

For four distinct primes in increasing order, the smallest possible set is $\{2, 3, 5, 7\}$, giving

$$\text{score} = 6 + 10 + 14 + 15 + 21 + 35 = 101,$$

which is not divisible by 3. For any other set of four primes, the largest prime is at least 11. The other three primes are at least $2, 3, 5$. Therefore, its score is at least the score for $\{2, 3, 5, 11\}$:

$$\text{score} = 6 + 10 + 22 + 15 + 33 + 55 = 141.$$

This score is divisible by 3 and is smaller than 288, so the smallest possible score is **(C)** 141.

13. There is a unique pair of positive integers (x, y) satisfying $x^2 - y^2 = 5x$. What is xy?

(A) 48 **(B)** 54 **(C)** 60 **(D)** 72 **(E)** 84

proposed by: Leo Wu

Solution: Rearrange as $x^2 - 5x = y^2$. Completing the square on the LHS, we obtain $(2x-5)^2 = 4y^2 + 25$. Rearrange again as $(2x-5)^2 - (2y)^2 = 25$; this is a difference of squares which factors as $(2x+2y-5)(2x-2y-5) = 25$.

Note that $x(x-5) = y^2 > 0 \implies x > 5$, so $2x+2y-5 > 0$. Therefore, $2x-2y-5 = 25/(2x+2y-5) > 0$. The divisors of 25 are 1, 5, and 25, so either 1) $2x+2y-5 = 2x-2y-5 = 5$ or 2) $2x+2y-5 = 25$ and $2x-2y-5 = 1$. The first case yields $y = 0$ which is not a positive integer, but solving the second case yields $x = 9$ and $y = 6$ so $xy = \boxed{\textbf{(B)}\ 54}$.

14. There are N distinct arrangements of the letters in MISSISSIPPI in which no two S's are adjacent. What is the sum of the digits of N?

(A) 5 **(B)** 6 **(C)** 7 **(D)** 10 **(E)** 15

proposed by: Prince Zhang

Solution: Using the four S's as dividers, let the lengths of the five blocks of non-S letters be a, b, c, d, e. Then $a+b+c+d+e = 7$, where $b, c, d \geq 1$ and $a, e \geq 0$. By stars and bars, there are $\binom{8}{4} = 70$ ways to choose these lengths. There are $\binom{7}{1,2,4} = 105$ ways to fill the 7 slots with I, M, P. Therefore, $N = 70 \cdot 105 = 7350$, so the sum of the digits of N is $\boxed{\textbf{(E)}\ 15}$.

15. What is the sum of the 199th powers of all 200 complex roots, counted with multiplicity, of

$$x^{200} + 15x - 5 = 0?$$

(A) -2985 **(B)** -2970 **(C)** -15 **(D)** 0 **(E)** 15

proposed by: Jonathan Wang

Solution: Let the roots be $r_1, r_2, \ldots, r_{200}$, counted with multiplicity. Since 0 is not a root, dividing $r_i^{200} + 15r_i - 5 = 0$ by r_i gives

$$r_i^{199} = \frac{5}{r_i} - 15.$$

By Vieta's formulas,

$$\sum_{i=1}^{200} \frac{1}{r_i} = \frac{r_2 r_3 \cdots r_{200} + \cdots + r_1 r_2 \cdots r_{199}}{r_1 r_2 \cdots r_{200}} = \frac{-15}{-5} = 3.$$

Therefore,

$$\begin{aligned}\sum_{i=1}^{200} r_i^{199} &= 5\sum_{i=1}^{200}\frac{1}{r_i} - 15\cdot 200\\ &= 5\cdot 3 - 3000\\ &= \boxed{\textbf{(A)}\ -2985}.\end{aligned}$$

16. An integer n satisfies $2 \le n \le 1000$. Exactly eight of the fractions

$$\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}$$

have prime denominators when written in lowest terms. How many possible values of n are there?

(A) 6 **(B)** 8 **(C)** 9 **(D)** 15 **(E)** 16

proposed by: Christopher Li

Solution: A prime p occurs as a reduced denominator only if p divides n. For each such prime, the fractions with reduced denominator p are exactly those equal to $1/p, 2/p, \ldots, (p-1)/p$. There are exactly $p-1$ such fractions, so

$$\sum_{\substack{p|n\\ p \text{ prime}}} (p-1) = 8.$$

Each prime divisor of n must therefore be at most 9, so the only primes that can divide n are $2, 3, 5, 7$, contributing $1, 2, 4, 6$, respectively. The only possible set of prime divisors is $\{3, 7\}$. Let $n = 3^a 7^b$ with $a, b \ge 1$.

- If $b = 1$, $n = 21, 63, 189, 567$.
- If $b = 2$, $n = 147, 441$.
- If $b \ge 3$, then $n \ge 3 \cdot 7^3 = 1029 > 1000$.

The number of possible values is therefore $4 + 2 = \boxed{\textbf{(A)}\ 6}$.

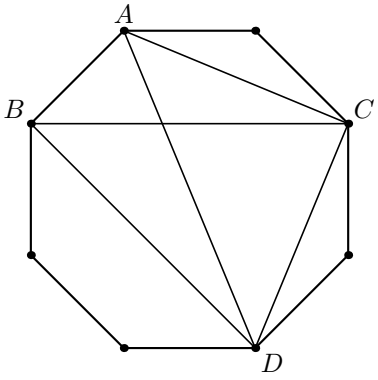
17. In triangle ABC, $\angle B = 45^\circ$ and $\angle A = 5\angle C$. Given that $AC = 2\sqrt{2}$, what is $AB^2 + BC^2$?

(A) 8 **(B)** 16 **(C)** $8 + 8\sqrt{2}$ **(D)** 32 **(E)** 64

proposed by: Leo Wu

Solution: One approach is to compute $\angle A = 112.5^\circ$ and $\angle C = 22.5^\circ$, then finish via the law of sines. A more slick approach is as follows.

Construct a regular octagon as shown in the diagram. Evidently $BC = BD$ and $AC = CD$, and we also have $\angle ABD = \angle ACD = 90^\circ$. Thus $AB^2 + BC^2 = AB^2 + BD^2 = AD^2 = 2AC^2 =$ $\boxed{\textbf{(B)}\ 16}$.



18. A rectangular grid consists of 4 rows and 12 columns of unit squares. A robotic ladybug travels from the bottom-left corner to the top-right corner, staying on the edges of these squares. Each move is one unit right, up, or down. The ladybug may never traverse the same unit edge twice.

    The ladybug chooses uniformly at random among all valid paths satisfying these conditions that reach the endpoint. Taking the bottom-left corner as the origin, let $P = (4, 2)$ and $Q = (8, 2)$. What is the probability that the ladybug's path passes through both P and Q?

    (A) $\frac{289}{625}$ **(B)** $\frac{17}{25}$ **(C)** $\frac{441}{625}$ **(D)** $\frac{561}{625}$ **(E)** $\frac{593}{625}$

    proposed by: Terry Yang

    Solution: Define a "12-point set" to consist of 12 points (i, h_i) for $i = 1, 2, \ldots, 12$, with $h_i \in \{0, 1, 2, 3, 4\}$. Given a path, record the point where the ladybug first reaches each vertical line $x = 1, 2, \ldots, 12$. These 12 points form a 12-point set. Since the ladybug never moves left and it never traverses the same unit edge twice, there is a bijection between 12-point sets and paths.

    The path passes through $P = (4, 2)$ if 2 is between h_4 and h_5, inclusive. The probability this happens is $1 - 2(2/5)^2 = 17/25$. Similarly, the probability the path passes through $Q = (8, 2)$ is $17/25$. Therefore, the probability the ladybug passes through both P and Q is

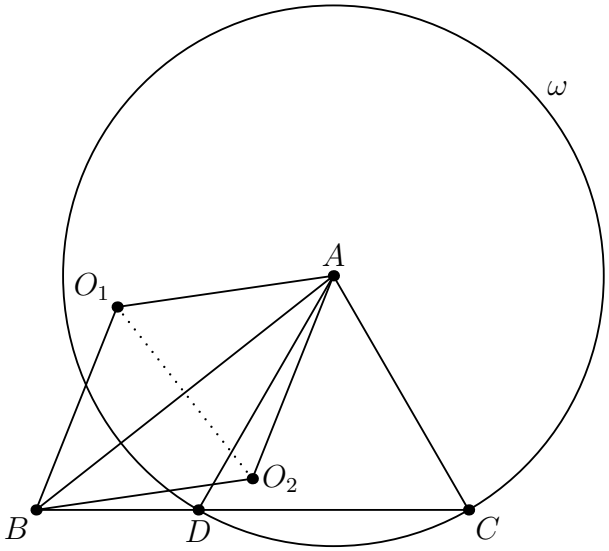
    $$\left(\frac{17}{25}\right)^2 = \boxed{\textbf{(A)}\ \frac{289}{625}}.$$

19. In triangle ABC, $AB = 14$ and $AC = 10$. Point D lies in the interior of side BC, with $AD = 10$ and $BD = 6$. Let O_1 and O_2 be the circumcenters of triangles ABD and ABC, respectively. The distance between O_1 and O_2 can be expressed as $m/\sqrt{n}$, where m, n are positive integers and n is not divisible by the square of any prime. What is mn?

(A) 14 **(B)** 17 **(C)** 21 **(D)** 42 **(E)** 84

proposed by: Lei He

Solution:



Let ω be the circle centered at A with radius $AD = 10$. The power of B with respect to ω is

$$BA^2 - AD^2 = BD \cdot BC$$
$$14^2 - 10^2 = 6(6 + DC).$$

Solving gives $DC = 10$. Therefore, $\triangle ADC$ is equilateral. Note that $\angle AO_1B = 2(180^\circ - \angle ADB) = 120^\circ$. Since $\triangle AO_1B$ is isosceles, $\angle O_1AB = \angle O_1BA = 30^\circ$. Similarly, $\angle AO_2B = 2\angle ACB = 120^\circ$. Since $\triangle AO_2B$ is isosceles, $\angle O_2AB = \angle O_2BA = 30^\circ$. Therefore, O_1 and O_2 are reflections of each other over AB. Let x be the distance from O_2 to AB. Then,

$$O_1O_2 = 2x = 2(7/\sqrt{3}) = 14/\sqrt{3},$$

giving $mn = \boxed{\textbf{(D)}\ 42}$.

20. For how many integers m with $1 \leq m \leq 100$ is there a nondegenerate triangle whose side lengths a, b, c satisfy

$$a^2 + b^2 + c^2 - \frac{1}{a+b+c} = ab + bc + ca$$

and

$$\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c} = m?$$

(A) 1 **(B)** 96 **(C)** 97 **(D)** 98 **(E)** 100

proposed by: Jonathan Wang

Solution: Let $f(a,b,c) = \dfrac{a}{b+c-a} + \dfrac{b}{c+a-b} + \dfrac{c}{a+b-c}$. Multiplying the first condition by $a+b+c$ gives

$$(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 1.$$

Note that this condition prevents $a=b=c$. Say a,b,c are the side lengths of a nonequilateral triangle. Since $a+b+c>0$ and $a^2+b^2+c^2-ab-bc-ca = \frac{(a-b)^2+(b-c)^2+(c-a)^2}{2} > 0$, we can write

$$(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = k^3$$

for some positive real k. This is equivalent to

$$\left(\frac{a}{k}+\frac{b}{k}+\frac{c}{k}\right)\left(\frac{a^2+b^2+c^2-ab-bc-ca}{k^2}\right) = 1.$$

Also, $f(a,b,c) = f(a/k, b/k, c/k)$. What this shows is that the set of all possible values of $f(a,b,c)$ satisfying the problem conditions is the same as the set of possible values of $f(a,b,c)$ with a,b,c side lengths of a nonequilateral triangle. We will calculate the latter set.

Using Ravi's substitution, say $a=x+y$, $b=y+z$, and $c=z+x$ $(x,y,z>0)$. Then,

$$f(x+y,y+z,z+x) = \frac{1}{2}\left(\frac{x+y}{z}+\frac{y+z}{x}+\frac{z+x}{y}\right) = \frac{1}{2}\left(\frac{x}{y}+\frac{y}{x}+\frac{y}{z}+\frac{z}{y}+\frac{z}{x}+\frac{x}{z}\right),$$

which is at least 3 by AM-GM. However, $f(x+y,y+z,z+x)=3$ is impossible since this would require $x=y=z$, giving $a=b=c$.

Taking $x=t$ and $y=z=1$ gives $f(x+y,y+z,z+x) = 1+t+1/t$. As t increases from 1 to ∞, this expression increases continuously from 3 to ∞. Therefore, every value greater than 3 is possible, giving $\boxed{\textbf{(C)}\ 97}$ possible values.

21. A machine initially displays 1. Each press of a button replaces the displayed integer n with

$$n + \gcd(n,17) + \operatorname{lcm}(n,17),$$

where gcd and lcm denote the greatest common divisor and least common multiple, respectively. For example, the first press changes 1 to 19. How many presses are needed for the displayed number to be divisible by 289 for the first time?

(A) 16 **(B)** 20 **(C)** 23 **(D)** 24 **(E)** 32

proposed by: Christopher Li

Solution: Let n_k be the displayed number after k presses. Then $n_0 = 1$. Since 17 is prime,

$$n + \gcd(n,17) + \operatorname{lcm}(n,17) = \begin{cases} 18n+1, & 17 \nmid n \\ 2n+17, & 17 \mid n \end{cases}.$$

For $0 \le k \le 16$, we have $n_k \equiv k+1 \pmod{17}$. Therefore, the recurrence for $0 \le k < 16$ is $n_{k+1} = 18n_k + 1$. We have

$$n_{16} = 1 + 18 + 18^2 + \cdots + 18^{16}.$$

By the binomial theorem, $18^k = (1+17)^k \equiv 1 + 17k \pmod{289}$. Therefore,

$$n_{16} \equiv 17 + 17(0 + 1 + \cdots + 16) \equiv 17 \pmod{289}.$$

Note that $17 \mid n_k \implies 17 \mid 2n_k + 17 = n_{k+1}$. For $k \ge 16$, let $n_k = 17q_k$. Then, $q_{16} \equiv 1 \pmod{17}$ and for $k \ge 16$,

$$\begin{aligned} n_{k+1} &= 2n_k + 17 \\ q_{k+1} &= 2q_k + 1 \\ q_{k+1} + 1 &= 2(q_k + 1). \end{aligned}$$

This implies

$$q_{16+k} + 1 = 2^k(q_{16} + 1) \equiv 2^{k+1} \pmod{17}.$$

Thus, $17 \mid q_{16+k} \iff 2^{k+1} \equiv 1 \pmod{17}$. Since

$$(2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7, 2^8) \equiv (2, 4, 8, 16, 15, 13, 9, 1) \pmod{17},$$

the smallest positive value of k satisfying $17 \mid q_{16+k}$ is 7. Therefore, the number of presses is $16 + 7 = \boxed{\textbf{(C) } 23}$.

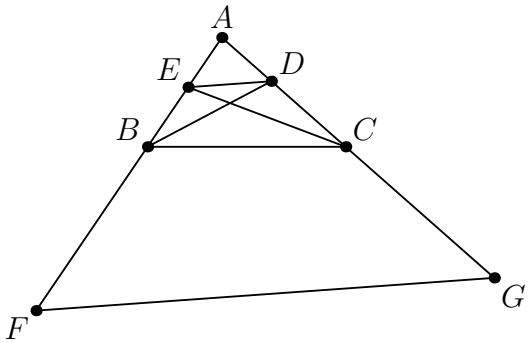
22. In triangle ABC, $AB = 4$, $AC = 5$, and $BC = 6$. The internal angle bisectors of $\angle B$ and $\angle C$ meet AC and AB at D and E, respectively. Extend AB beyond B to F and AC beyond C to G so that $BF = BC = CG$.

What is the ratio $\frac{ED}{FG}$?

(A) $\frac{4}{121}$ **(B)** $\frac{2}{11}$ **(C)** $\frac{1}{5}$ **(D)** $\frac{2}{5}$ **(E)** $\frac{5}{11}$

proposed by: Avneet Prakash

Solution:



By angle bisector theorem,

$$\frac{AD}{DC} = \frac{AB}{BC} = \frac{2}{3}, \qquad \frac{AE}{EB} = \frac{AC}{BC} = \frac{5}{6}.$$

Since $AC = 5$ and $AB = 4$, it follows that $AD = 2$ and $AE = 20/11$. We also have $AF = AB + BF = 10$ and $AG = AC + CG = 11$. Thus,

$$\frac{AE}{AF} = \frac{20/11}{10} = \frac{2}{11} = \frac{AD}{AG}.$$

This implies $ED \parallel FG$, so $\frac{ED}{FG} = \frac{AE}{AF} = \boxed{\textbf{(B)}\ \frac{2}{11}}$.

23. Arcane has five cats weighing 2, 4, 8, 16, and 32 kilograms. She places them on a seesaw one at a time, choosing a remaining cat and a side on each turn. After every placement, the total weight on the left side must exceed the total weight on the right side. Two sequences are considered different if the order of the cats or the side chosen for any cat differs. How many such sequences place all five cats on the seesaw?

(A) 945 **(B)** 960 **(C)** 1890 **(D)** 1920 **(E)** 3840

proposed by: Lei He

Solution: Let a_n be the number of valid sequences for n cats weighing $2, 2^2, \ldots, 2^n$.

Note that the left side is heavier exactly when the heaviest cat currently on the seesaw is on the left. Given a valid sequence for n cats, we do casework on when the lightest cat is placed.

Case 1: The lightest cat is placed first

This cat must be placed on the left. Divide all cats weights by 2. There are a_{n-1} ways to place the $n-1$ remaining cats. Therefore, there are a_{n-1} sequences of length n.

Case 2: The lightest cat is not placed first

The lightest cat can be placed on either side, since it will not be the heaviest cat on the seesaw. There are $2(n-1)$ ways to place the lightest cat, and a_{n-1} ways to place the other $n-1$ cats. Therefore, there are $2(n-1)a_{n-1}$ sequences of length n.

Combining the two cases gives

$$a_n = (2n-1)a_{n-1}.$$

Since $a_1 = 1$, $a_5 = 9 \cdot 7 \cdot 5 \cdot 3 \cdot 1 = \boxed{\textbf{(A)}\ 945}$.

24. Consider the sum

$$S = \sum_{k=2}^{2027} k^2(k+1)!.$$

What is the largest positive integer n such that $n!$ divides S?

(A) 2026 **(B)** 2027 **(C)** 2028 **(D)** 2029 **(E)** 2030

proposed by: Avneet Prakash

Solution: We would like this sum to telescope, so let

$$k^2(k+1)! = a_k(k+1)! - a_{k-1}k!.$$

Dividing by $k!$ gives us

$$k^3 + k^2 = a_k(k+1) - a_{k-1}.$$

Now suppose that a_k is a polynomial of degree 2:

$$a_k = Ak^2 + Bk + C.$$

Substituting this expression gives

$$k^3 + k^2 = Ak^3 + Bk^2 + (2A + C)k + B - A.$$

Matching coefficients, we obtain $a_k = k^2 + k - 2$. Thus,

$$\sum_{k=2}^{n} k^2(k+1)! = \sum_{k=2}^{n} \left[(k+1)!(k^2+k-2) - k!\left((k-1)^2 + (k-1) - 2\right)\right].$$

Telescoping the series gives us

$$\begin{aligned}\sum_{k=2}^{n} k^2(k+1)! &= (n+1)!(n^2+n-2) \\ &= (n+2)!(n-1).\end{aligned}$$

Substituting $n = 2027$ gives $2026 \cdot 2029!$, so $\boxed{\textbf{(D)}\ 2029}$ is the answer.

25. An arrow is drawn inside each square of a 2×6 rectangular grid of squares, pointing up, down, left, or right. A square is "pointed to" if an arrow in a square sharing a side with it points towards it. How many ways are there to draw the arrows so that every square is pointed to?

(A) 13 **(B)** 121 **(C)** 144 **(D)** 156 **(E)** 169

proposed by: Leo Wu

Solution: There are 12 arrows and 12 squares so each square is pointed to exactly once.

Colour the grid in a white and black checkerboard pattern. Since white squares only point to black squares and black squares only point to white squares, we can consider these two types of arrows separately.

For each arrow in a white square and the black square it points to, merge these two squares into a 1×2 domino. These six dominoes must tile the grid, since each arrow points to a distinct black square.

Conversely, for any tiling of the grid with dominos, the above process can be reversed to obtain six arrows in white squares pointing to distinct black squares. We have established a bijection between valid arrow configurations on white squares and domino tilings!

In general, there are F_{n+1} ways to tile a $2 \times n$ grid with dominos, where F_n is the nth Fibonacci number. This can be shown by induction; the base cases $n = 1, 2$ are easily verified. If the rightmost domino is vertical, the remaining $2 \times (n-1)$ grid can be tiled in F_n ways. If the rightmost two dominos are horizontal, the remaining $2 \times (n-2)$ grid can be tiled in F_{n-1} ways. Summing yields F_{n+1} as claimed.

Here $n = 6$ so there are $F_7 = 13$ ways to draw the arrows in white squares. Analogously, there are also 13 ways to draw the arrows in black squares, for a final answer of $13^2 = \boxed{\textbf{(E)}\ 169}$.